

JEE MAIN REVISION MASTER: ENVIRONMENTAL CHEMISTRY 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 5 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Kcet Environmental Chemistry - Core Application	5 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Environmental Chemistry - Core Application (Moderate Yield — 5 PYQs)

Core Principle: Key problem pattern in Kcet Environmental Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The first chlorinated organic insecticide prepared is..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "Identify the incorrect statement from the following...." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "Which of the following is not a green house gas?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

Identify the incorrect statement from the following.

- (A) Oxides of nitrogen in the atmosphere can cause depletion of the ozone layer.
- (B) Ozone absorbs the intense ultraviolet radiation of Sun.
- (C) Depletion of ozone layer is because of its chemical reactions with chlorofluoro alkanes.
- (D) Ozone absorbs infrared radiation.

Correct Answer: None

Solution Analysis: Option D is incorrect. Ozone primarily absorbs ultraviolet (UV) radiation, particularly in the wavelength range of 240 to 320 nanometers, which helps to protect living organisms from harmful UV rays emitted by the sun. It does not absorb infrared radiation significantly; instead, gases such as carbon dioxide (CO_2), water vapor (H_2O), and methane (CH_4) are primarily responsible for the absorption of infrared radiation, contributing to the greenhouse effect.